

Long Ordinal - Data imputation

Setup

```
library(here)
library(tidyverse)
library(mice)
library(miceadds)
library(arrow)
library(data.table)
```

Load data

These are historic data from patients of the Division of Medical Oncology at USB, which had their first appointment from 2023 to mid 2025, i.e., before LIQPLAT started. The data are not publicly available.

```
set.seed(1234)
df_raw <- read_parquet(here("data", "long_data.parquet"))
```

Simplify some baseline variables

```
# remove observations after day 182
# lump factor variables according to frequency to have a maximum of 5 levels

df <- df_raw |>
  select(-c(randomselect_liqplat, offerpart_liqplat, icf_reg, q29, q30, QL2_trans,
PF2_trans, PF2_trans_imput, diagnosis_details, y, leukocytes, lymphocytes_relative)) |>
  mutate(
    diagnosis = fct_lump_n(diagnosis, n = 4),
    diagnosis = factor(as.integer(diagnosis)),
    ecog_fstcnt = factor(ecog_fstcnt, levels = 1:3, ordered = TRUE),
    in_hospital = case_when(
      in_hospital == 1 ~ TRUE,
      TRUE ~ FALSE),
    unplanned_stay = case_when(
      unplanned_stay == 1 ~ TRUE,
      TRUE ~ FALSE),
    er_visit = case_when(
      er_visit == 1 ~ TRUE,
      TRUE ~ FALSE)
  ) |>
  filter(day < 183) |>
  group_by(pat_id) |>
  mutate(
    n_blood_products_all = sum(as.numeric(n_blood_products), na.rm = T),
    survival_time = max(day),
    died = if_else(any(death == 1), 1, 0)
  ) |>
  arrange(pat_id, day) |>
```

```
ungroup() |>
select(-n_blood_products)
```

Add Nelson-Aalen-estimates, “Randomize”

Nelson-Aalen yields slightly better predictions than raw survival time, see: <https://stefvanbuuren.name/fimd/sec-toomany.html>

```
na_df <- df |>
  select(pat_id, survival_time, died) |>
  unique()

# nelson aalen estimator
na_est <- cbind(na_df$pat_id, nelsonaalen(na_df, timevar = survival_time, statusvar =
died)) |>
  data.table() |>
  rename(pat_id = V1, na_est = V2)

df <- df |>
  left_join(na_est, by = "pat_id")

# create random group allocation.
tx <- sample(unique(df$pat_id), size = 0.5 * length(unique(df$pat_id)), replace =
FALSE)

df <- df |>
  mutate(tx = if_else(pat_id %in% tx, "1", "0"))
```

Variables for final analysis

When analyzing the trial data, we will use the following predictors for imputation and modeling:

Demographics and baseline clinical:

- age - age at enrollment
- gender - patient sex
- ecog_fstcnt - ECOG performance status (ordered factor: 0-4)
- diagnosis - primary cancer diagnosis (lumped to 5 most frequent categories)

Laboratory values:

- albumin - serum albumin (g/L)
- c_reactive_protein - CRP (mg/L)
- bilirubin - serum bilirubin (μmol/L)
- lactate_dehydrogenase - LDH (U/L)

Healthcare utilization:

- n_blood_products_all - number of blood product units received over 182 days

Clinical events:

- event_next_treatment - indicator for starting next line of treatment
- event_progression - indicator for disease progression

Outcome tracking:

- died - death indicator whether patient died within 182 days
- na_est - Nelson-Aalen cumulative hazard estimate
- tx - treatment group allocation

Longitudinal outcome:

- day - day after random selection
- in_hospital - indicator for being in hospital on that day
- er_visit - indicator for emergency room visit on that day

Some variables are currently not available in the dummy dataset, so the code on this list might not match.

Multiple imputation

Baseline values will be imputed at the patient (cluster level).

```
md.pattern(df)
```

```
init <- mice(df, maxit = 0)
pred_matrix <- init$predictorMatrix
meth <- init$method

# don't want to use survival time because we use na_est.
pred_matrix[, "survival_time"] <- 0 # 'time' will not predict any other variable
pred_matrix["survival_time", ] <- 0 # 'time' will not be predicted by any other
variable

# Use patient ID as clustering variable
pred_matrix[, "pat_id"] <- -2

# use mean matching at the cluster level for baseline observations
# this ensures only one ecog per patient per imputation
meth["age"] <- "2lonly.pmm"
meth["diagnosis"] <- "2lonly.pmm"
meth["gender"] <- "2lonly.pmm"
meth["ecog_fstcnt"] <- "2lonly.pmm"
meth["albumin"] <- "2lonly.pmm"
meth["bilirubin"] <- "2lonly.pmm"
meth["c_reactive_protein"] <- "2lonly.pmm"
meth["lactate_dehydrogenase"] <- "2lonly.pmm"
meth["n_blood_products_all"] <- "2lonly.pmm"
meth["died"] <- "2lonly.pmm"
meth["na_est"] <- "2lonly.pmm"

# These are not missing in our dummy data but might have some missingness in the trial
data
meth["in_hospital"] <- "2l.pmm"
meth["unplanned_stay"] <- "2l.pmm"
meth["er_visit"] <- "2l.pmm"

imputed_data <- mice(df,
```

```

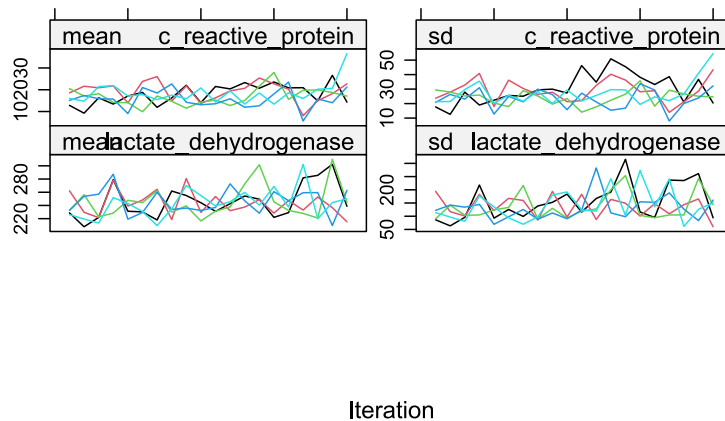
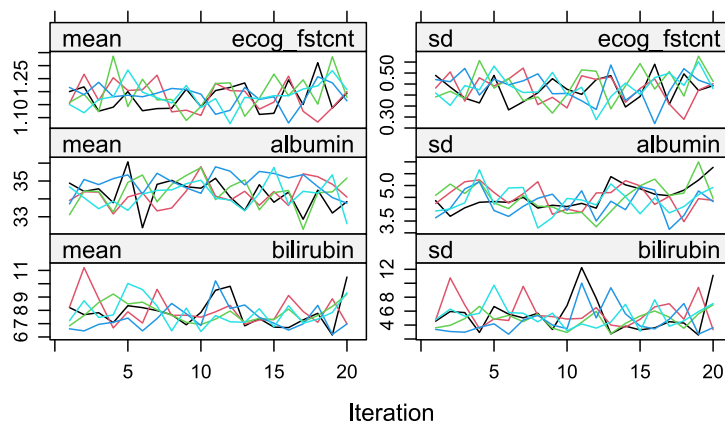
m = 5, # will set to 50
maxit = 20, #will set to 50
predictorMatrix = pred_matrix,
method = meth,
seed = 1234)

```

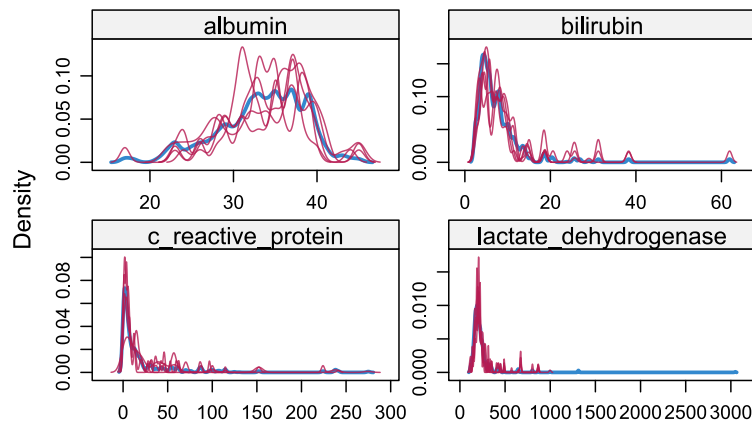
Diagnostics

The diagnostics look fine to me.

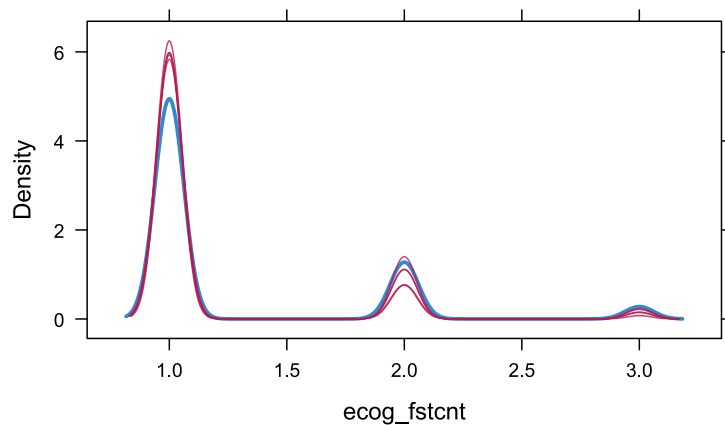
```
plot(imputed_data)
```



```
densityplot(imputed_data)
```



```
densityplot(imputed_data, ~ ecog_fstcnt)
```



Select one imputation

For the analysis of the trial data we'd of course use all the imputed datasets and stack the posterior draws. However, for our simulations this is unfeasible, and we shall thus only use one complete dataset for the setup. I will use the data imputed in long format for the time being.

```
complete_data <- complete(imputed_data, 4)
```

Export

```
write_parquet(complete_data, sink = here("data", "hosp_long_complete.parquet"))
```